

2023

JOURNEYMAN

OPEN

BOOK

FINAL EXAM #2



70 QUESTIONS

TIME LIMIT - 3 HOURS

TIME SPENT **MINUTES**

SCORE **%**

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1. Portable structures for fairs, carnivals and similar events shall not be located under or within a space 15 feet horizontally of conductors operating in excess of ____ volts.

- (a) 1000 (b) 600 (c) 300 (d) 250

2. Which of the following wiring methods is **not** permitted in the ceiling space used as a return-air plenum?

- (a) PVC (b) MI cable (c) AC cable (d) IMC conduit

3. A copper-coated steel ground rod shall be at least ____.

- (a) 3/8" (b) 1/2" (c) 9/16" (d) 5/8"

4. Shore power for boats shall be provided by single receptacles rated not less than ____ amperes.

- (a) 15 (b) 20 (c) 25 (d) 30

5. Where a rock bottom is encountered, the ground rod shall be permitted to be buried in a trench that is at least ____ deep.

- (a) 2 1/2 feet (b) 3 feet (c) 24" (d) 48"

6. A ____ is a portion of a sign that may provide protection from the weather but is not an electrical enclosure.

- (a) Section Sign (b) Sign Body (c) Sign Gutter (d) Sign Enclosure

7. Internal parts of electrical equipment, including busbars, wiring terminals, insulators, and other surfaces, shall not be damaged or contaminated by foreign materials such as paint, plaster, cleaners, abrasives, or ____.

- (a) moisture (b) organic residues (c) corrosive residues (d) dust and dirt

8. Where a building or structure has any combination of feeders, branch circuits, or services passing through it or supplying it, a permanent plaque or directory shall be installed at each feeder and branch circuit ____ location denoting all other services, feeders, or branch circuits supplying that building or structure or passing through that building or structure and the area served by each.

- (a) disconnect (b) connection (c) termination (d) entrance

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9. _____. The highest level that water can reach before it spills out.

- (a) Over Flow (b) Escape Limit (c) Datum Level (d) Maximum Water Level**

10. The _____ number of branch circuits shall be determined from the total calculated load and the size or rating of the circuits used.

- (a) minimum (b) maximum (c) total (d) largest**

11. An open resistor when checked with an ohmmeter reads _____.

- (a) infinite (b) zero (c) low but not zero (d) high but within the tolerance**

12. In relation to a transformer, the ratio 20:1 indicates that _____.

- (a) there are 20 turns in the primary and 1 turn in the secondary
(b) secondary voltage is 1/20 of the primary voltage
(c) for every 20 turns on the primary, there is one turn on the secondary
(d) all of these**

13. In a parallel circuit, the voltage across each branch is _____ the source voltage.

- (a) equal to (b) greater than (c) less than (d) none of these**

14. When a 60 watt bulb is connected in series with a 500 watt room heater and the bulb is replaced with a 25 watt bulb on a 120 volt circuit the _____.

- (a) heater output will decrease
(b) heater output will increase
(c) bulb will not glow
(d) heater output will remain unchanged**

15. If the secondary voltage of a transformer is step-down, the primary will have _____.

- (a) half as many turns as the secondary (b) fewer turns than the secondary
(c) more turns than the secondary (d) as many turns as the secondary**

16. In a parallel circuit with an unequal resistance on each branch _____.

- (a) the power drawn on each branch is equal
(b) the current on each branch is equal
(c) the voltage across each branch is equal
(d) none of these**

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17. A circuit with a lagging current means the circuit is ____.

- (a) capacitive (b) reactive (c) inductive (d) at resonance

18. The ____ value is considered as the most important value of a sine wave.

- (a) peak (b) instantaneous (c) effective (d) average

19. A wire has a resistance of 5 ohms. What will be the resistance of another wire of the same material three times as long and half the cross sectional area?

- (a) 7.5Ω (b) 15Ω (c) 30Ω (d) 50Ω

20. Flexible cord used in extension cords made with separately listed and installed components shall be permitted to be supplied by a branch circuit in accordance with ____.

- (a) sized per Table 400.5(A)(1)&(2)
(b) sized per Table 310.15(B)(16)
(c) separate components are not allowed to be field assembled
(d) 20 amp circuits - #16 AWG and larger

21. Warning signs shall be ____ posted at points of access to conductors in all conduit systems and cable systems. The warning sign(s) shall carry the following wording: Danger-High Voltage-Keep Out.

- (a) permanently (b) clearly (c) legibly (d) conspicuously

22. The grounding electrode conductor shall be of copper, aluminum, copper-clad aluminum, or the items permitted in 250.68(C). The material selected shall be resistant to any ____ condition existing at the installation or shall be protected against corrosion.

- (a) chemical (b) corrosive (c) deteriorating (d) damaging

23. One type of enclosure permitted for use in outdoor corrosive environments is ____.

- (a) Type 4 X (b) Type 6 (c) Type 12 (d) Type 13

24. Table ____ lists the volume allowances required per conductor for outlet, device and junction boxes.

- (a) 314.16(A) (b) 310.60 (c) 314.16(B) (d) 300.5

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25. Overhead spans of ____ conductors and open multiconductor cables shall have a vertical clearance of not less than 8' 6" above the roof surface.

(a) insulated (b) open (c) service (d) bare

26. Nonconductive coatings (such as paint, lacquer, and enamel) on equipment to be grounded shall be removed from threads and other contact surfaces to ensure good electrical continuity by means of fittings ____ so as to make such removal unnecessary.

(a) approved (b) labeled (c) listed (d) designed

27. ____ lists the dimensions of insulated conductors and fixture wires.

(a) 300.1(C) (b) 300.19(A) (c) Table 2, Chapter 9 (d) Table 5, Chapter 9

28. Table ____ lists demand factors for kitchen equipment other than dwelling units.

(a) 220.55 (b) 220.56 (c) 220.83 (d) 220.84

29. ____ lists the percent of cross sectional fill permitted in conduit and tubing fill for conductors and cables.

(a) Table 4, Chapter 9 (b) Table 8, Chapter 9
(c) Table 1, Chapter 9 (d) Table 5, Chapter 9

30. Signs and outline lighting systems with lampholders for incandescent lamps shall be marked to indicate the maximum allowable lamp wattage per lampholder. The markings shall be permanently installed, in letters at least ____ high, and shall be located where visible during relamping.

(a) 1/4" (b) 1/2" (c) 1" (d) 2"

31. ____ is the process by which one conductor produces or induces a voltage in another conductor even though there is no mechanical coupling between the two conductors.

(a) Cutting of fluxes (b) Short circuit (c) Induction (d) Eddy current

32. Electrical current is measured in terms of ____.

(a) electron pressure (b) electrons passing a point per second
(c) watts (d) resistance

33. The resistance of a circuit may vary due to ____.

(a) a loose connection (b) change in voltage (c) change in current (d) induction

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34. An alternation is ____.

- (a) one-half cycle (b) one hertz (c) one alternator (d) two cycles

35. A shunt is used to measure ____.

- (a) resistance (b) capacitance (c) current (d) wattage

Half way through the exam_____

36. If two equal resistance conductors are connected in parallel, the resistance of the two conductors is equal to ____.

- (a) the resistance of one conductor
(b) twice the resistance of one conductor
(c) one-half the resistance of one conductor
(d) the resistance of both conductors

37. Table ____ lists the maximum rating or setting of motor branch-circuit short circuit and ground fault protective devices.

- (a) 430.248 (b) 430.52(C)(1) (c) 430.37 (d) 430.22

38. Branch circuits recognized by this article shall be rated in accordance with the maximum permitted ampere rating or ____ of the overcurrent device.

- (a) listing (b) marking (c) setting (d) rating

39. Table ____ lists the maximum cord-and-plug-connected load to receptacle.

- (a) 210.21(B)(2) (b) 210.24 (c) 210.21(B)(3) (d) 220.3

40. Table ____ lists clearances over roadways, walkways, rail, water, and open land.

- (a) 300.5 (b) 220.102 (c) 235.360(A) (d) 225.61

41. Which of the following is **not** considered part of a luminaire?

- (a) ballast (b) a lampholder (c) lamp or lamps (d) parts designed to position the light source

42. ____ RMC shall be permitted to be installed for direct burial and swimming pool applications.

- (a) Stainless steel (b) Galvanized steel (c) Aluminum (d) Red brass

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43. Backfill that contains large rocks, paving materials, cinders, large or sharply angular substances, or corrosive material shall not be placed in an excavation where materials may damage raceways, cables, conductors, or other structures or prevent adequate compaction of fill or contribute to the ____ of raceways, cables, or other substructures.

- (a) damage (b) deterioration (c) aging (d) corrosion**

44. Enclosures that are not over 100 cubic inches in size and which have two conduits supported within three feet on either side of the enclosure and the enclosure does not contain devices or support fixtures shall not be required to have the enclosure supported if the conduits are ____.

- (a) rigid nonmetallic conduits
(b) threaded into hubs identified for the purpose
(c) installed with locknuts inside and outside enclosure
(d) shoulders of fittings outside and locknuts inside the box**

45. Conductors drawn from a copper-clad aluminum rod with the copper metallurgically bonded to an aluminum core. The copper forms a minimum of ____ percent of the csa.

- (a) 10 (b) 20 (c) 40 (d) 70**

46. Concrete, brick or tile walls are considered as being ____.

- (a) isolated (b) insulators (c) grounded (d) dry locations**

47. An isolated ground receptacle shall be identified by ____.

- (a) CO/ALR orange marking on the face of the receptacle
(b) a metal faceplate not less than 0.030 inches in thickness
(c) a reset type test button on the face of the receptacle
(d) an orange triangle located on the face of the receptacle**

48. In a hospital General Care Area, each patient bed location shall be provided with a minimum of ____ receptacles.

- (a) 3 single (b) 2 single or 1 duplex (c) 2 duplex or 4 single (d) 4 duplex or 8 single**

49. Balconies, decks, and porches that are attached to the dwelling unit and are accessible from inside the dwelling unit shall have at least one receptacle outlet accessible from the balcony, deck, or porch. The receptacle outlet shall not be located more than ____ above the balcony, deck, or porch walking surface.

- (a) 6' (b) 6' 6" (c) 7' (d) 8'**

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50. Conductors having ____ insulation and operating at different voltage levels shall not occupy the same enclosure, cable, or raceway.

(a) **thermoset** (b) **flame-retardant** (c) **nonshielded** (d) **silicone rubber**

51. The total opposition to current flow in an AC circuit is expressed in ohms and is called ____.

(a) **impedance** (b) **conductance** (c) **reluctance** (d) **resistance**

52. If a 240 volt heater is used on 120 volts, the amount of heat produced will be ____.

(a) **twice as great** (b) **four times as great** (c) **1/4 as much** (d) **the same**

53. The length of time that a fault current would flow on the equipment grounding conductor would be approximately ____.

(a) **1/2 cycle** (b) **6 seconds** (c) **60 cycles** (d) **120 cycles**

54. A one-quarter bend in a raceway is equivalent to an angle of ____ degrees.

(a) **90** (b) **45** (c) **25** (d) **180**

55. Which of the following will **not** affect the resistance of a circuit?

(a) **Length of the Conductor** (b) **Diameter of the Conductor**
(c) **Insulation of the Conductor** (d) **Temperature**

56. The voltage will lead the current when the ____ in the circuit.

(a) **inductive reactance exceeds the capacitive reactance**
(b) **reactance exceeds the resistance in the circuit**
(c) **resistance exceeds reactance**
(d) **capacitive reactance exceeds the inductive reactance**

57. The decimal equivalent for 11/16" is ____.

(a) **.8125** (b) **.6875** (c) **.5625** (d) **.9375**

58. Extreme ____ may cause PVC conduit to become brittle, and therefore more susceptible to damage from physical contact.

(a) **heat** (b) **cold** (c) **sunlight** (d) **moisture**

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59. Where metal fences are located within ____ feet of the exposed electrical conductors or equipment, the fence shall be bonded to the grounding electrode system with bonding jumpers.

- (a) 25 (b) 24 (c) 20 (d) 16**

60. In multifamily dwellings, Type NM Cable is permitted in buildings that are permitted to be type(s) ____.

- (a) V (b) III (c) IV (d) all of these**

61. When sizing a branch circuit for a fixed storage-type water heater with a capacity of 120 gallons or less, the water heater is considered a/an ____.

- (a) varying load (b) intermittent load (c) noncontinuous load (d) continuous load**

62. In other than one-and two-family dwelling units, the available fault current and the date of the calculation was performed shall be field marked on the enclosure of ____.

- (a) switchgear (b) panelboards (c) switchboards (d) all of these**

63. If the emergency disconnecting means required in 230.85 for dwelling units is a meter disconnect, it must be ____ the meter mounting equipment.

- (a) near (b) remote from (c) adjacent to (d) integral to**

64. If locating a motor's disconnecting means within sight of the motor would result in an increased hazard, the motor controller may act as the required disconnect if it is ____.

- (a) serviceable (b) accessible (c) lockable (d) readily accessible**

65. A three-way switch is counted as ____ conductor(s) based on the largest conductor connected to the switch in determining conductor fill in a device box.

- (a) one (b) two (c) three (d) not required to be counted**

66. Equipment intended to interrupt current at fault levels shall have a/an ____ rating, at normal voltage, that is at least equal to the available fault current at the line terminals of the equipment.

- (a) time delay (b) interrupting (c) short-time (d) long-time**

67. Vegetation such as trees shall **not** be used for support of ____.

- (a) luminaires (b) boxes (c) overhead conductor spans (d) any of these**

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68. A luminaire in a commercial cooking hood must, among other requirements, exclude grease, oil, and ____ from the lamp and wiring compartment.

(a) solids (b) moisture (c) oxygen (d) exhaust vapors

69. Stainless steel rigid metal conduit may use galvanized steel boxes and enclosures if those enclosures are not subject to ____.

(a) severe corrosive influences (b) unauthorized contact
(c) corrosive influences (d) physical damage

70. The maximum size FMT permitted is ____.

(a) 1/2" (b) 3/4" (c) 1" (d) 1 1/4"
